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Potential black hole seeding of the bulgeless spiral galaxy NGC 4424 via an infalling star cluster

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ABSTRACT

Galaxies can grow through their mutual gravitational attraction and subsequent union. While orbiting a regular high-surface-brightness galaxy, the body of a low-mass galaxy can be stripped away. However, the stellar heart of the infalling galaxy, if represented by a tightly bound nuclear star cluster, is more resilient. From archival *Hubble Space Telescope* images, we have discovered a red, tidally-stretched star cluster positioned 4.″9 (∼390 pc in projection) from, and pointing towards the center of, the post-merger spiral galaxy NGC 4424. The star cluster, which we refer to as 'Nikhuli', has a near-infrared luminosity of $6.6^{+2.0}_{-1.6} \times 10^6\ L_{\odot,F160W}$ and likely represents the nucleus of a captured/wedded low-mass early-type galaxy. Moreover, from our *Chandra X-ray Observatory* images, Nikhuli is seen to contain a high-energy X-ray point source unlikely to be either a low- or high-mass X-ray binary. It has a photon index $\Gamma = 1.7$ and an X-ray luminosity $L_{0.5-8\,\mathrm{keV}} = 6.31^{+7.50}_{-3.77} \times 10^{38}$ erg s^{-1} (90% confidence). Lacking an outward-pointing comet-like appearance, the stellar structure of Nikhuli favours infall rather than the ejection from a gravitational-wave recoil event. A minor merger with an early-type galaxy may sew a small classical bulge and sow a massive black hole. The stellar mass and the velocity dispersion of NGC 4424 predict a central black hole of (0.6–1.0)×10^5 M$_\odot$, similar to the expected intermediate-mass black hole in Nikhuli, and suggestive of a black hole supply mechanism for bulgeless late-type galaxies such as NGC 4424. We may potentially be witnessing black hole seeding by capture and sinking, with a nuclear star cluster the delivery vehicle.

1. INTRODUCTION

Galaxies have long been known to merge. Smaller galaxies rain down on bigger galaxies, stoking their growth (Lindblad 1927; Spitzer & Baade 1951; Baade & Minkowski 1954; Vorontsov-Velyaminov 1959; Ibata et al. 2019b). This evolution has become a defining tenet of the cold dark matter model. The ongoing discovery of (infalling) satellites continues to test and support the hierarchical growth scenario for galaxies. The burgeoning detection of drawn-out, low surface brightness, debris streams of former galaxies in the outskirts of larger galaxies (Putman et al. 1998; Belokurov et al. 2006; Martin et al. 2014; Koposov et al. 2015; Drlica-Wagner et al. 2015; Shipp et al. 2018; Martinez-Delgado et al. 2021) are starting to reveal the extent of the hierarchical model, at least in regard to building stellar halos. The closest known tidal stream to the center of our galaxy is 10-20 kpc distant from the galaxy center (Yuan et al. 2020; Malhan et al. 2021).

The dense and compact nuclear star clusters — which are common features at the centers of low (≈10^7–10^9 M$_\odot$) and intermediate stellar mass (≈10^9–10$^{10.5}$ M$_\odot$) galaxies(e.g. Sandage et al. 1985; Böker et al. 2002; Graham & Guzmán 2003; Balcells et al. 2003, 2007; Scott & Graham 2013; Sánchez-Janssen et al. 2019) — are more strongly gravitationally bound than their host galaxies and are thus more able to resist disruption during the cannibalistic process (e.g., Bekki et al. 2001, 2003). Indeed, the dual and multiple nuclei of massive early-type galaxies is a testament to the robustness of these captured star clusters during the galaxy assimilation process (e.g., Jenner 1974; Petrosian et al. 1978; Koroviakovskii et al. 1981; Tonry 1985; Blakeslee & Tonry 1992; Bonfini & Graham 2016). Furthermore, the abundance of kinematically distinct cores and nuclear stellar disks in early-type galaxies (e.g. Franx et al. 1989; Forbes et al. 1995; Graham et al. 1998; Hau et al. 1999) reveals

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a more complete journey to the center of these galaxies, in stark contrast with an eternity spent in the halo.

Such galaxy growth, revealed by off-center nuclei, can be harder to spot in late-type galaxies due to confusion with knots of star formation. This ambiguity makes the later stages of the merger/digestion process more challenging to establish in late-type galaxies. Furthermore, without a massive stellar bulge, dynamical friction will be less in the inner regions, reducing the extent to which minor mergers contribute to the nuclei and bulges. The abundance of low bulge-to-total stellar mass ratios in late-type galaxies identified by Graham & Worley (2008) and Weinzirl et al. (2009) was later used by Kormendy et al. (2010) to challenge the hierarchical growth of most late-type galaxies. It was speculated that the bulges were too small to be built by mergers and were thus considered the result of internal secular processes. The detection of a infalling nuclear star cluster near the center of a late-type spiral galaxy would reveal that the Roche-lobe destruction of galactic neighbors does not just produce streams that build diffuse halos, but that tidal-shredding also contributes to the galaxy proper, including small bulges. Moreover, such digestion could potentially seed and feed spiral galaxies with massive black holes.

Among the Virgo cluster's spiral galaxies, the SB(s)a galaxy NGC 4424 is of particular interest. While there is no detectable X-ray point source at the center of NGC 4424, there is a nearby ($5''$) 'off-center' source, NGC 4424 X-3 ($L_{0.5-8\,\mathrm{keV}} \approx 6 \times 10^{38}$ erg s^{-1}). First publicly reported by Boselli et al. (2018), they suggested it is either an X-ray binary (XRB) or the nucleus of a merged galaxy. Two brighter X-ray point sources plus a fainter fourth point source reside in the outskirts of this galaxy.

Although Feng & Soria (2011), Bachetti et al. (2014) and Kaaret et al. (2017) reveal why super-Eddington accretion (Alexander & Natarajan 2014; Soria et al. 2014) is thought to explain many of the off-center ultraluminous X-ray sources (ULX: $L_X \approx 3 \times 10^{39}$ to 10^{41} erg s^{-1}), some might be intermediate-mass black holes (IMBHs) from accreted galaxies, particularly the hyperluminous X-ray sources (HLXs: $L_X > 10^{41}$ erg s^{-1}, Gao et al. 2003) such as ESO 243-49 (Farrell et al. 2009; Soria et al. 2010) and others (Sutton et al. 2012; Barrows et al. 2019; Earnshaw et al. 2019). Here, we report the association of NGC 4424 X-3 with a red, elongated star cluster.

In Section 2, we provide some background information and references to NGC 4424 before presenting optical and near-infrared images along with an X-ray map. We perform a re-analysis of the *Hubble Space Telescope* (*HST*) image, the *Chandra X-ray Observatory CXO* and XMM-Newton data, and provide an extended discussion. Section 3 offers a prediction for the mass of the central black hole in both NGC 4424 and the suspected infaller. Finally, a more detailed discussion, centered around off-centered sources, is provided in Section 4.

Previously, in Graham et al. (2019), we used the mean redshift-independent distance modulus for NGC 4424 of 30.6±1.0 from the NASA/IPAC Extragalactic Database (NED)[1]. Here, we adopt the Cepheid-based distance modulus of 31.080±0.292 (16.4±0.8 Mpc) from Riess et al. (2016). This encompasses and agrees with an array of recent results, including Hatt et al. (2018), who report a 'tip of the red giant branch' distance modulus of $31.00 \pm 0.03_{\mathrm{stat}} \pm 0.06_{\mathrm{sys}}$ ($15.8 \pm 0.2_{\mathrm{stat}} \pm 0.4_{\mathrm{sys}}$ Mpc), and with Munari et al. (2013), who derived a SN Ia distance modulus of 30.95, and with Cortés et al. (2008), who reported a value of 30.91 based on the Tully & Fisher (1977) relation.

2. OBSERVATIONS AND FINDINGS

2.1. *Preface*

From deep *CXO* exposures of 100 early-type galaxies in the Virgo cluster — half of which possess a nuclear star cluster (Côté et al. 2006; Ferrarese et al. 2006) — Gallo et al. (2010) effectively reveals that nuclear star clusters, in and of themselves, tend not to contain X-ray point sources. Among the 30 lowest mass galaxies of that 100-strong sample, most are nucleated, and all have been predicted to harbour an IMBH. Graham & Soria (2019) reported that just three of these 30 contained a central X-ray point source, ruling out stellar-mass XRBs as a common feature of nuclear star clusters, at least in early-type galaxies. If these star clusters are comprised of both a young and old stellar population, it will rule out both high- and low-mass XRBs, respectively.

From past observations, NGC 4424 (Figure 1) has been identified as a post-merger galaxy undergoing substantial central star formation (Kenney et al. 1996). Cortés et al. (2006) and Boselli et al. (2018) provide a detailed analysis of NGC 4424, finding that it experienced an unequal-mass merger event less than 0.5 Gyr ago. The encounter has fueled an outburst of ionized gas, evident as a ∼10 kpc plume pointing in the opposite direction to a ∼110 kpc long (ram pressure)-stripped tail of HI gas emanating from NGC 4424 as it plows through the hot intracluster medium of the Virgo cluster.

2.2. *Optical/near-IR images*

NGC 4424 appears as a bulgeless galaxy containing a knotty bar-like feature extending to a radius of $\approx 14''$ (with a brighter inner region extending to a radius of $\approx 7''$) plus an exponential-like disk. For our investigation, we have obtained an archival Wide Field Camera 3 Infrared (WFC3/IR) F160W H-band image taken as a part of *Hubble Space Telescope* (*HST*) Proposal ID 12880 (PI: A.Riess). The near-IR helps to reduce the impact

[1] http://nedwww.ipac.caltech.edu

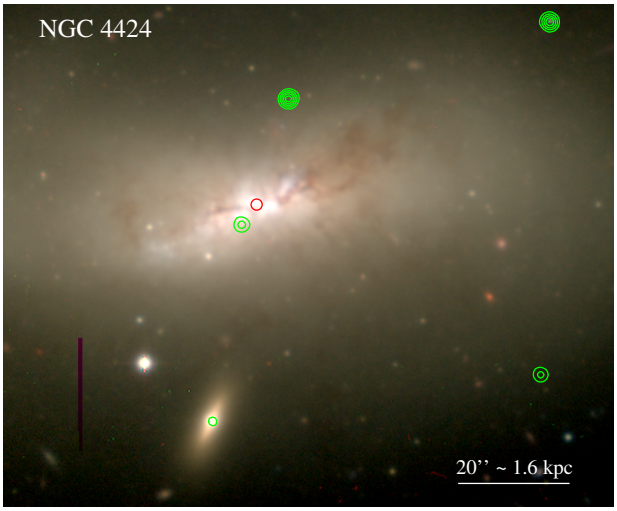

Figure 1. Next Generation Virgo Cluster Survey (NGVS: Ferrarese et al. 2012) image of the post-merger galaxy NGC 4424 (red = i filter; green = g; blue = u), with *CXO*/ACIS-S contours (0.5–7.0 keV band) overlaid in green. North is up, east is to the left. The red circle shows the NED-provided position for the galaxy's optical nucleus, with a radius of 1 arcsec capturing the associated uncertainty. The near-central, but off-center, X-ray point source (NGC 4424 X-3) is associated with an elongated star cluster (see Figure 2). The galaxy to the south is IC 3366, a background galaxy at a distance of ∼110 Mpc, while the vertical red stripe is an artifact that should be ignored.

of dust, young stars and avoid H_α emission. The image, shown in Figure 2, has a Point Spread Function (PSF) Full Width at Half Maximum (FWHM) spatial resolution of 0.″18. Our data reduction process followed that detailed in Davis et al. (2019).

We modeled NGC 4424 using the Isofit/Cmodel task (Ciambur 2015), run within the Image Reduction and Analysis Facility (IRAF)[2] package ELLIPSE. The model consists of a series of one-dimensional profiles capturing the radial run of intensity, major-axis position angle, ellipticity and a sequence of higher-order Fourier Harmonic terms capturing the isophotes' departures from a pure ellipse. The resulting two-dimensional galaxy model — built from the assorted one-dimensional profiles — has been subtracted from the image to highlight the residual structure. This has revealed an elongated, off-center and off-bar star cluster 4.″9 arcseconds (390 pc) to the southeast of the galaxy center. This is exactly where the off-center X-ray point source resides (Figure 1, see also Boselli et al. (2018)). This find is reminiscent of the discovery of the optical counterpart/host of HLX-1 in the galaxy ESO 243-49 (Soria et al. 2010), which can be clearly seen in Figure 7 of (Ciambur 2015). Not surprisingly, the residual image for both ESO 243-49 and NGC 4424 also more clearly display additional (point source)-like knots associated with dusty star formation in their disk plane, plus probably also some globular clusters.

The somewhat stretched star cluster in NGC 4424 is drawn out in the direction of galaxy's center and likely represents the nucleus of a captured galaxy undergoing the final digestion stages of merging and assimilation with NGC 4424. As the likely seed of a galaxy harvested from the nearby environment and consumed by NGC 4424, we refer to this star system as Nikhuli[3], which is a name relating to a festive period of celebrating and wishing for a rich harvest.

From the WFC3/IR F160W residual image (galaxy minus model), the elongated source has a flux of 844±29 electrons per second. Using a WFC3/IR F160W zero-point[4] of 25.946 mag (AB photometric system: Koekemoer et al. 2013), this corresponds to an apparent magnitude of 18.63±0.04 mag (AB). Using a distance modulus of 31.08±0.29 gives an absolute magnitude of −12.45±0.29 mag. With an F160W absolute magnitude for the Sun of +4.60 mag (AB mag: Willmer 2018), the stellar luminosity is $6.6^{+2.0}_{-1.6} \times 10^6\, L_{\odot,F160W}$. Assuming an F160W stellar mass-to-light ratio of 0.5 (Graham & Spitler 2009, see their Figure A1), the stellar mass is $3.3 \times 10^6\, M_\odot$.

Nikhuli also shows up in F814W imagery, ruling out the possibility that the detection in the F160W image was solely due to FeII 1.64 μm line emission from an ionized bubble around NGC 4424 X-3. The elongated morphology also disfavours such an interpretation. We used Gaussian smoothing to degrade the spatial resolution of the WFC3 F814W image — obtained from the same *HST* Proposal (ID 12880. PI: A.Riess) as the F160W image — to match the 2.33 pixel seeing in the F160W image. The (galaxy plus star cluster) light in a 3.″4 (272 pc) by 1.″5 (120 pc) ellipse centered on the star cluster has a mean F814W−F160W color equal to 1.60±0.10 mag. This color is slightly redder than the surrounding mean galaxy color of 1.44±0.04 mag, and may be indicative of an older stellar population. This would not be unusual. For example, the star cluster in the Milky Way dwarf spheroidal satellite Eridanus II is also thought to be old (Alzate et al. 2021). While the star cluster in NGC 4424 is not a blue knot of recent star formation, it could be dust-reddened.

[2] http://ast.noao.edu/data/software

[3] Taken from the Sumi language, spoken by a quarter of a million (formerly all tribal) people in Nagaland, the name relates to the Tuluni, *aka* Anni, festival.

[4] The sensitivity of the WFC3/IR detector has been remarkably stable over the past decade (Kozhurina-Platais & Baggett 2020) and offers a 1σ photometric accuracy/repeatability of 1.5 per cent (0.015 mag) (Bajaj 2019).

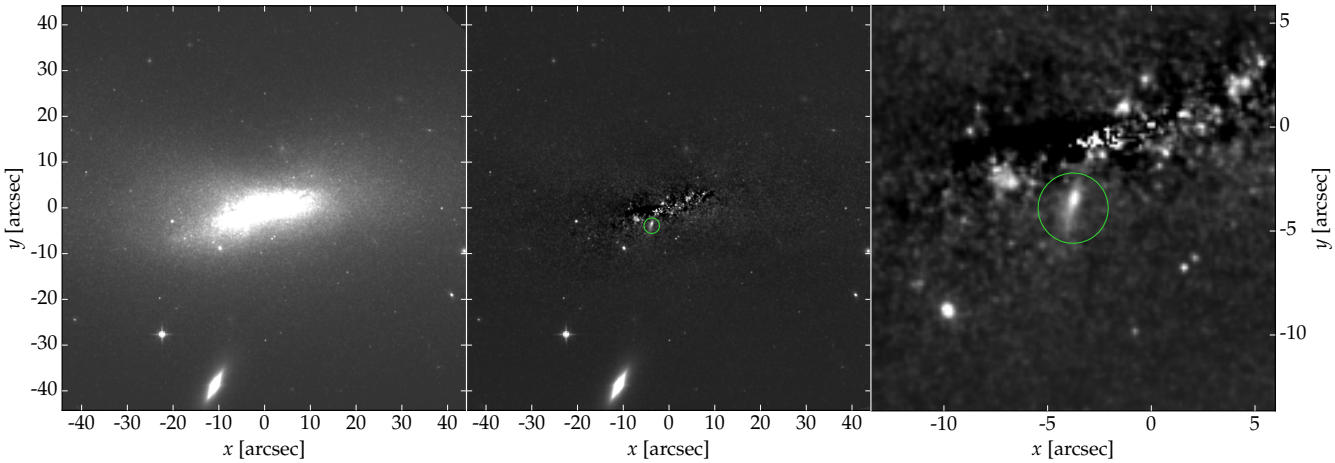

Figure 2. NGC 4424: *HST* WFC3/IR F160W cut-out image (left panel: $88'' \times 88'' \approx 7\text{kpc} \times 7\text{kpc}$); Isofit residual image (middle panel); zoom-in of the region containing NGC 4424 X-3 (right panel).

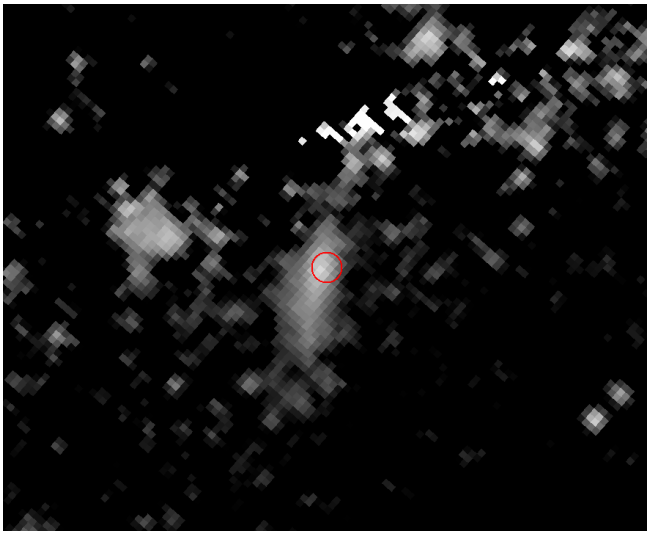

Figure 3. Residual image from Figure 2, with a slightly different contrast and with the center of the X-ray point-source NGC 4424 X-3 overlaid. The red circle has a radius of $0.''3$, and it encapsulates the 90% uncertainty on the X-ray source position after mapping into the *HST* image.

Because the (color of the) galaxy light can dominate over the (color of the) star cluster light, the F814W−F160W galaxy (plus star cluster) color image is not ideal for determining the color of Nikhuli nor for seeing faint substructures. We have therefore attempted to derive a residual color image rather than a galaxy color image. We subtracted the F160W residual image from the F814W residual image. By design, most of the pixels will have intensities close to zero. As such, the average color in the residual color image should be close to zero rather than equal to the galaxy color. Furthermore, roughly half of the pixels in the F814W and F160W residual images will have a negative intensity. This can be problematic because the color, given by the ratio of intensities, $-2.5\log(I_1/I_2)$, is ill-defined when I_1/I_2 is negative, and the color image is noisy where either I_1 or I_2 are close to zero. To mitigate against this, in the residual images we added a pixel intensity of $+1$ to those pixels with a positive intensity, and we have subtracted a pixel intensity of -1 to those pixels with a negative intensity. For each residual image, we then calculated the pseudo-magnitudes $-2.5\log(I+1)$ for those pixels with positive values of I, and $+2.5\log(|I-1|)$ for those pixels with negative values of I. We then used $\text{mag}_1-\text{mag}_2$ as our proxy for each pixels color. The prominent stripe seen in the residual color image (Figure 4) is from the WFC3 chip gap in the F814W image. Unfortunately, due to the imperfect processes of image alignment, seeing matching, pixel-size matching, plus the addition of the 'bias level' to the intensity, this pseudocolor image for the residual structure is indicative rather than quantitative. While the cyan and orange colors may be noise, the darker blue and red colors should be meaningful.

2.3. *X-ray data*

2.3.1. *Chandra X-ray Observatory data*

Using the 14.87 ks *CXO* Advanced CCD Imaging Spectrometer (ACIS) exposure (ObsID: 19408. PI: Soria) taken on 2017-04-17, Boselli et al. (2018, see their sections 4.2) reported on the obscured X-ray source NGC 4424 X-3, located $4''.9$ (≈390 pc) to the south-east of the nucleus. The *CXO* provides accurate sub-arcsecond localizations of faint point-like sources and NGC 4424 X-3 is coincident with the star cluster.[5] Figure 3 shows the X-ray point-source at

[5] The *CXO/HST* alignment was refined with the help of the point-like *CXO* source associated with the nucleus of the background galaxy at R.A.=12:27:12.16, Dec.=+9:24:35.55 and with a quasar at R.A.=12:27:12.85 Dec.=+9:24:13.13.

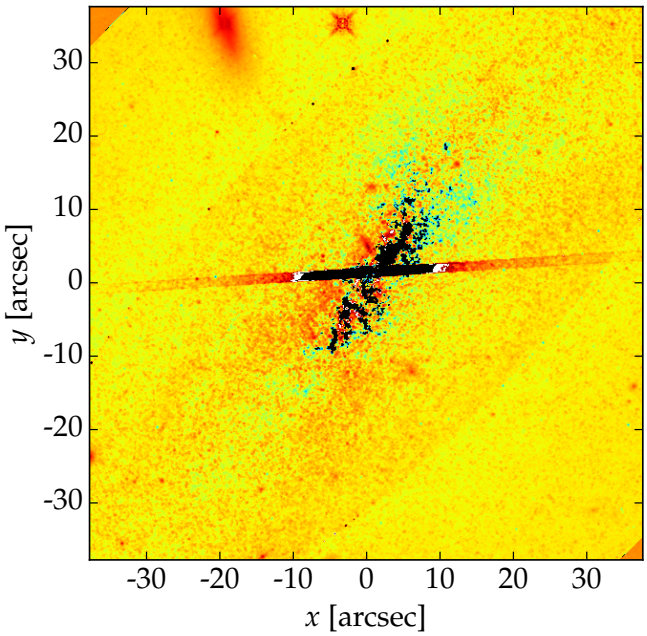

Figure 4. Residual color image after removing the mirror-symmetric component of NGC 4424 about its major-axis. The F160W residual image (Figure 2) has been subtracted from the F814W residual image (not shown). North is to the lower-right. Yellow denotes a near absence of flux and thus a neutral color. Residual flux shows up as red or blue depending on the relative intensity in the F160W and F814W residual images. Light blue and orange can be considered noise. See section 2.2 for details.

R.A.=12:27:11.801, Dec.=+9:25:10.71, with a red circle denoting the 90% radius = 0.3″.

We have reanalyzed the *CXO* data, following the detailed prescription provided in Graham et al. (2021) and Soria et al. (2021). Briefly, we used the Chandra Interactive Analysis of Observations (CIAO) Version 4.12 software package (Fruscione et al. 2006) with the Calibration Database Version 4.9.1, to reprocess the event files (CIAO task *chandra_repro*), and the display package DS9 (Joye & Mandel 2003) for image processing, centroiding, and selection of source and background regions. NGC 4424 X-3 is located close to the ACIS aimpoint, which makes the point spread function sharper and reduces the positional uncertainty of the centroid even with a small number of counts. To take advantage of this, we also extracted images at sub-pixel resolution (0.5 pixels, corresponding to ≈0″.25), with *dmcopy*, and used the 0.3–7 keV sub-pixel image for centroiding.

We used a source extraction radius of 1″.5 (3 pixels) around the source centroid; the expected encircled energy within this radius is ≈95%. For the background, we used the annulus between 2″.5 and 10″. There are 8 X-ray counts within the source region (in fact, all within 1″.0 of the centroid position), in the 0.3–7 keV band: none are in the soft band (0.3–1 keV), 1 is in the medium band (1–2 keV), and 7 are in the hard band (2–7 keV). Splitting the energy band in a different way, we see that all 8 counts are >1.5 keV. Although the number of counts is small, the detection is highly significant, because the background is very low; based on the local average, we expect only ≈0.1 background counts in the soft band, in the source region; ≈0.2 background counts in the medium band; and ≈0.1 background counts in the hard band. For $N = 8$ detected counts and ≈0.4 expected background counts, the 90% Bayesian confidence interval is ≈4–13 net counts, and the 99% interval is ≈2–17 net counts (Kraft et al. 1991). The net 0.3–7 keV count rate (including the PSF correction to an infinite radius) is $0.56^{+0.42}_{-0.29} \times 10^{-3}$ ct s^{-1} (90% Bayesian confidence interval).

We tried to infer more information on the flux and luminosity of X-3, in two different ways. First, we applied the *srcflux* task to the reprocessed event file to obtain a model-independent estimate of the observed, (H_I gas)-dimmed, flux[6] in the "broad" ACIS band (0.5–7 keV). Second, despite the small number of counts, we extracted a spectrum and its associated response and ancillary response files with *specextract*. We regrouped the spectrum to 1 count per bin, and fitted it with simple models (power-law and disk-blackbody) using the XSPEC (Arnaud 1996) package version 12.11.0. We used the Cash statistics (Cash 1979) for the the fit (*cstat* in XSPEC).

The main purpose of our spectral analysis was obviously not to determine a complex fitting model, but at least to estimate what intrinsic column density and de-absorbed luminosity are consistent with the peculiar energy distribution of the few observed photons (all above 1.5 keV). Assuming a power-law model (the simplest option in the absence of further information) with a photon index $\Gamma = 1.7$ (a standard, reference choice for the hard emission of accreting compact objects, e.g. Molina et al. 2009; Corral et al. 2011; Younes et al. 2011; Plotkin et al. 2013; Yang et al. 2015), we estimate that an intrinsic absorbing column $N_{\rm H,int} \gtrsim 10^{22}$ cm^{-2} (much higher than the Galactic line-of-sight column density $N_{\rm H,Gal} = 1.5 \times 10^{20}$ cm^{-2}; (HI4PI Collaboration et al. 2016)) is needed to explain the energy distribution of the detected photons (Table 1); a similarly high absorption is required if the spectral model is a disk-blackbody with an assumed temperature of 1.0 keV. This rough estimate of the intrinsic absorption permits a more realistic conversion of count rate and flux to

[6] *srcflux*'s model-independent flux is based on the rate and energies of the detected photons, taking into account the quantum efficiency and effective area within the source region; instead, the model-dependent flux is the flux of the best-fitting model. They are alternative approximations to the "true" observable (absorbed) flux. From the absorbed model-dependent flux, *srcflux* also computes the unabsorbed flux, which is used to estimate the emitted luminosity.

intrinsic luminosity. For the assumed[7] $\Gamma = 1.7$ power-law model[8], the unabsorbed 0.3–10 keV luminosity is $L_{0.3-10} \approx 10^{+14}_{-6} \times 10^{38}$ erg s^{-1}. For the disk-blackbody model, $L_{0.5-8} \approx 9^{+14}_{-5} \times 10^{38}$ erg s^{-1} (Table 1). As a further check of these results, we input the net count rates in the hard band (derived earlier with *srcflux*) into the CIAO version of the Portable, Interactive Multi-Mission Simulator (PIMMS) version 4.11a[9], with response functions for *Chandra* Cycle 18. We recovered the same estimates for the intrinsic flux (corresponding to an extrapolated 0.3–10 keV luminosity of about 10^{39} erg s^{-1} for plausible spectral slopes), and the same high values of $N_{\rm H,int} \gtrsim 10^{22}$ cm^{-2} to remove all photons below $\approx$1.5 keV. The *CXO* image is shown in Figure 1.

We do not have enough counts and bandwidth coverage to decide between a power-law model and a curved (e.g., disk-blackbody) model, even with the Cash statistics. A power-law model is the expected spectral shape for an IMBH, which would be in the low/hard state at a luminosity of $\approx 10^{39}$ erg s^{-1} ($\sim 10^{-3}$ times the Eddington luminosity for a $10^4 M_\odot$ BH). However, the luminosity of $\approx 10^{39}$ erg s^{-1} is also consistent with a stellar-mass BH at the top of the high/soft state, with a typical temperature of $\approx$1 keV and an inner disk radius $\sim$50–100 km.

2.3.2. *XMM-Newton data*

The field around NGC 4424 X-3 was also observed with *XMM-Netwon* European Photon Imaging Camera - Metal Oxide Semi-conductor (EPIC-MOS) on two occasions: on 2010 June 13 (Obs.ID: 0651790101; Revolution: 1925) and 2017 December 5 (Obs.ID: 0802580201; Revolution: 3295). We downloaded the data from the HEASARC archive and reprocessed them with the Science Analysis Software SAS version 17.0.0; we rebuilt event files for the pn and MOS cameras with *epproc* and *emproc*, respectively. We filtered the event files to remove intervals of high or rapidly flaring particle background; we selected only intervals with a background RATE parameter <0.35 for MOS1 and MOS2, and <0.5 for the pn, at channels 10000 < PI < 12000. After this filtering step, the good-time-interval for Obs.ID 0651790101 in the source extraction region was 13.8 ks for the pn, and 20.6 ks for each of the two MOS cameras; for Obs.ID 0802580201, the live time was 11.1 ks for the pn and 18.9 ks for the MOSs. We then filtered the event

[7] The power-law slope, aka photon index, Γ, is around 1.7 for Eddington ratios $\approx 10^{-5}$–10^{-2} (e.g. Molina et al. 2009; Corral et al. 2011; Younes et al. 2011; Plotkin et al. 2013; Yang et al. 2015). For Eddington ratios less than 10^{-5}, Γ steepens to around 2.1, and for Eddington ratios greater than 10^{-2}, Γ steepens to around 2.5.

[8] With $\Gamma = 1.7$, one has that $L_{0.5-8\,\rm keV} = 1.075\,L_{0.5-7\,\rm keV}$, and $L_{0.5-10\,\rm keV} = 1.207\,L_{0.5-7\,\rm keV}$. Similarly, $L_{2-10\,\rm keV} = 0.725\,L_{0.5-8\,\rm keV}$, and $L_{0.5-10\,\rm keV} = 1.123\,L_{0.5-8\,\rm keV}$.

[9] https://cxc.harvard.edu/toolkit/pimms.jsp

Table 1. Best-fitting parameters of the *Chandra*/ACIS spectrum of NGC 4424 X-3. Error ranges are 90% confidence limits for 1 interesting parameter. Values in brackets were frozen in the fit.

Model Parameters	Values
Model-independent 0.5–7 keV flux from *srcflux*	
Net Rate (10^{-3} ct s^{-1})	$0.56^{+0.42}_{-0.29}$
$f_{0.5-7}$ (10^{-14} erg cm^{-2} s^{-1})	$0.79^{+0.59}_{-0.40}$
phabs × *phabs* × *powerlaw*	
$N_{\rm H,Gal}$ (10^{22} cm^{-2})	[0.015]
$N_{\rm H,int}$ (10^{22} cm^{-2})	$2.7^{+3.8}_{-1.9}$
Γ	[1.7]
$N_{\rm po}$ (10^{-6} photons keV^{-1} cm^{-2} s^{-1} at 1 keV)	$4.5^{+6.1}_{-2.8}$
C-stat/dof	2.9/7
$f_{0.5-7}$ (10^{-14} erg cm^{-2} s^{-1})^a	$1.1^{+1.0}_{-0.6}$
$L_{0.3-10}$ (10^{38} erg s^{-1})^b	$10.0^{+13.7}_{-6.1}$
phabs × *phabs* × *diskbb*	
$N_{\rm H,Gal}$ (10^{22} cm^{-2})	[0.015]
$N_{\rm H,int}$ (10^{22} cm^{-2})	$3.4^{+4.6}_{-2.3}$
$kT_{\rm in}$ (keV)	[1.0]
$N_{\rm dbb}$ (10^{-3} km^2)^c	$1.3^{+2.1}_{-0.8}$
$R_{\rm in}\sqrt{\cos\theta}$ (km)^d	70^{+43}_{-28}
C-stat/dof	3.4/9
$f_{0.5-7}$ (10^{-14} erg cm^{-2} s^{-1})^a	$0.81^{+0.69}_{-0.42}$
$L_{0.3-10}$ (10^{38} erg s^{-1})^b	$8.5^{+13.5}_{-5.4}$

^{a}: absorbed model flux in the 0.5–7 keV band.
^{b}: isotropic de-absorbed luminosity in the 0.3–10 keV band, defined as $4\pi d^2$ times the de-absorbed 0.3–10 keV model flux.
^{c}: $N_{\rm dbb} = (r_{\rm in}/d_{10})^2 \cos\theta$, where $r_{\rm in}$ is the apparent inner disk radius in km, d_{10} the distance to the source in units of 10 kpc (here, $d_{10} = 1640$), and θ is our viewing angle ($\theta = 0$ is face-on).
^{d}: $R_{\rm in} \approx 1.19 r_{\rm in}$ for a standard disk (Kubota et al. 1998).

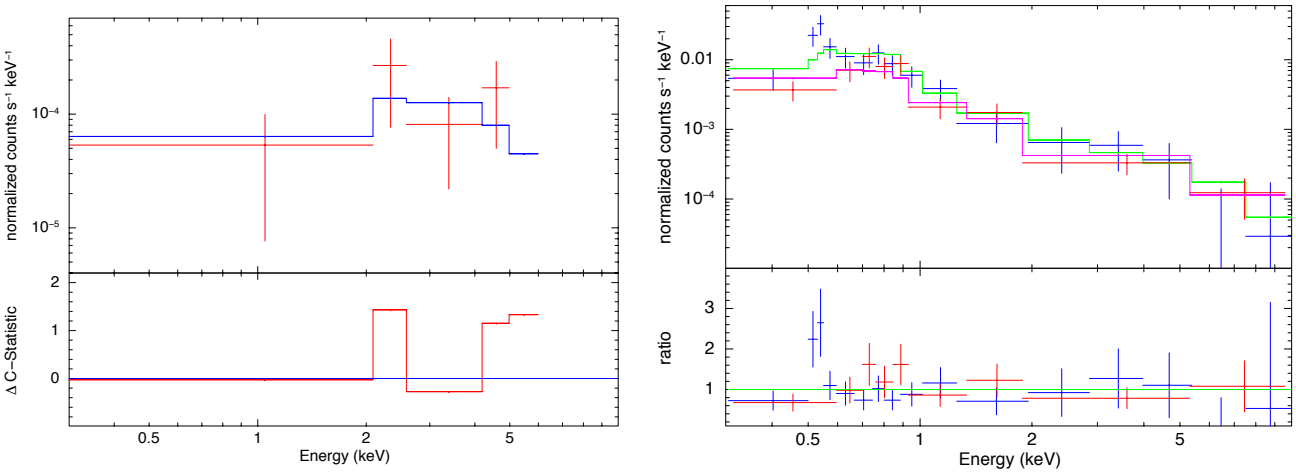

Figure 5. Left panel: CXO/ACIS-S spectrum of NGC 4424 X-3 in Nikhuli, with a $\Gamma = 1.7$ power-law model (blue line). The data points have been grouped to a signal-to-noise >1.8 for plotting purposes only. The fit was done on the individual counts, using Cash statistics. Right panel: Combined *XMM-Newton*/EPIC spectra of NGC 4424 X-3 in 2010 and 2017, along with the best-fitting models and the data/model ratios. Red data points are for the 2010 spectrum; the corresponding best-fitting model is plotted in magenta. Blue data points are for the 2017 spectrum; the corresponding best-fitting model is plotted in green. The EPIC spectra were grouped to 1 point per bin and fitted with the Cash statistics; the results were later rebinned to a signal-to-noise ratio >2.5 and a maximum number of grouped channels = 50, for plotting purposes only. The photon index (locked between the two epochs) of the power-law component is $\Gamma = 1.7^{+0.9}_{-0.7}$; the de-absorbed 0.3–10 keV luminosity of the power-law (i.e., not including the thermal plasma emission) is $\approx 10^{39}$ erg s^{-1} in the 2010 EPIC spectrum and $\approx 2.2 \times 10^{39}$ erg s^{-1} in the 2017 spectrum.

files with the standard conditions "(FLAG==0) && (PATTERN<=4)" for the pn, and "(#XMMEA_EM && (PATTERN<=12)" for the MOSs (corresponding to single and double events).

We defined a circular source region of radius 15″: this is slightly smaller than usual for EPIC point-like sources, but we wanted to reduce contamination of the faint point-like source from the surrounding diffuse emission expected from the hot gas in the central region of this star-forming galaxy. (This is obviously not an issue in *Chandra*, given its much higher spectral resolution). We defined local background regions seven times larger than the source region. For each of the two observations, we used *xmmselect* to extract individual spectra for the pn and the MOSs; we built the associated response and ancillary response files with *rmfgen* and *arfgen*. Finally, we combined the pn and MOS spectra of each observation with *epicspeccombine*. We grouped the two resulting EPIC spectra to a minimum of 1 count per bin, and fitted them with XSPEC (Arnaud 1996) version 12.11.0, using the Cash statistics (Cash 1979). We chose the Cash statistics because the combined spectra do not have enough counts for a meaningful χ^2 fitting: we estimate ≈190 net counts in the EPIC spectrum from Obs.ID 0651790101, and ≈120 net counts from Obs.ID 0802580201.

Both spectra are well fitted with thermal plasma emission, dominating below ≈1 keV, and a power-law component, dominating at higher energies. In particular, the soft component is consistent with multi-temperature thermal plasma. When fitted with a two *apec* components, the two temperatures are ≈0.2 keV and ≈0.7 keV. This is the characteristic temperature distribution of diffuse hot gas in star-forming galaxies (Owen & Warwick 2009; Mineo et al. 2012). Thus, we interpret the power-law component as the emission from the accreting compact object, and the soft component as a local enhancement of diffuse emission, tracer of recent star formation (perhaps associated with the satellite merger). The soft thermal component must be seen through a much lower column density than estimated from the hard colors of the *Chandra* detection.

Based on this tentative interpretation, and in order to limit the number of free parameters, we froze the temperatures of the two *apec* components at 0.2 keV and 0.7 keV, and added an additional intrinsic absorbing column density of 10^{22} cm^{-2}, seen only by the power-law component. We also kept the *mekal* normalizations and the power-law photon index locked between the two epochs. We obtain a good fit (C-statistics of 496.8/497) for a photon index $\Gamma = 1.7^{+0.9}_{-0.7}$. The 0.3–10 keV deabsorbed luminosity of the power-law component (which corresponds to the point-like *Chandra* source, in our model interpretation) roughly doubled from $\approx 10^{39}$ erg s^{-1} in 2010 to $\approx 2.2 \times 10^{39}$ erg s^{-1} in 2017 (0.5–8 keV luminosities of $\approx 8 \times 10^{38}$ erg s^{-1} and $\approx 1.8 \times 10^{39}$ erg s^{-1}, respectively).

In passing we note that because the X-ray source has not faded over 7 years, it is unlikely to have originated from a stellar tidal disruption event (TDE) from one of

the stars in the star cluster approaching too close to the putative black hole (Baumgardt et al. 2006; Komossa 2015; Lin et al. 2018).

3. BLACK HOLE MASS ESTIMATES

NGC 4424 might harbour two massive black holes, possibly with one yet to settle to the center of this merged system (Merritt & Milosavljević 2005; Barausse 2012; Tremmel et al. 2018), thereby explaining NGC 4424 X-3 in the elongated, off-center star cluster. In this scenario, the second (speculated) massive black hole is quiescent, residing at the center of NGC 4424. Here, we first report on the predicted mass for such a galaxy-centric black hole. We then provide a prediction for the mass of the black hole in the star cluster Nikhuli.

3.1. *NGC 4424*

Using a revised distance modulus of 31.080±0.292 (= 16.4 ± 0.8 Mpc), rather than 30.6±1.0, NGC 4424 has a revised (total) stellar mass $\log M_{*,\mathrm{gal}} = 10.0 \pm 0.1$ (Graham et al. 2019, see their Appendix). This mass is based on the near-IR K'-band (2.2 μm) galaxy magnitude of 8.86 mag from GOLD Mine[10] (Gavazzi et al. 2003), plus our assumed 0.10 mag uncertainty and a K'-band stellar mass-to-light ratio for the galaxy of $M/L_{K'} = 0.62 \pm 0.09$. This galaxy stellar mass implies a central black hole mass of $\log M_{\mathrm{bh}} = 4.8 \pm 0.8$ (based upon a symmetrical Bayesian analysis of the M_{bh}–$M_{*,\mathrm{gal}}$ relation for late-type galaxies) or 5.1 ± 0.8 (based upon a symmetrical bisector regression for late-type galaxies using the modified FITEXY routine). See Davis et al. (2018) and Graham et al. (2019) for details.

The merger event in NGC 4424 may have elevated the central stellar velocity dispersion, reported to be 57.0 ± 8.6 km s^{-1} by Ho et al. (2009). Although, this is perhaps in line with the (merger-induced increase to the) stellar mass as it results in a consistent prediction for the black hole mass of $\log M_{\mathrm{bh}} = 5.0 \pm 0.8$ (Davis et al. 2017). The past/ongoing merger has, however, distorted the spiral pattern (Kenney et al. 1996; Cortés et al. 2006), likely accounting for the discrepant prediction, see Table 2, for the black hole mass obtained using the spiral arm pitch angle (16.9±2.4 degrees: Graham et al. 2019) and the M_{bh}–ϕ relation in Davis et al. (2017).

The error-weighted mean of the logarithm of the estimated black hole masses can be calculated via

$$\overline{\log M_{\mathrm{bh}}} = \frac{\sum_{i=1}^{N} w_i \log M_{\mathrm{bh},i}}{\sum_{i=1}^{N} w_i}, \tag{1}$$

with inverse-variance weighting[11] such that $w_i = 1/(\delta \log M_{\mathrm{bh},i})^2$. The associated 1σ standard error is given by

$$\delta\, \overline{\log M_{\mathrm{bh}}} = \sqrt{1/\sum_{i=1}^{N} w_i}. \tag{2}$$

Based on the stellar mass and velocity dispersion of NGC 4424, it is predicted to possess a central intermediate mass black hole with $\overline{\log M_{\mathrm{bh}}} = 4.9 \pm 0.6$ dex. The benefit of such an error-weighted mean is that the uncertainty is less than the near-order of magnitude uncertainty on the individual black hole mass estimates.

Table 2. Black hole mass predictions for the center of NGC 4424.

$\log M_{\mathrm{bh}}$ ($M_{*,\mathrm{total}}$)	$\log M_{\mathrm{bh}}$ (ϕ)	$\log M_{\mathrm{bh}}$ (σ)	$\overline{\log M_{\mathrm{bh}}}$
4.8±0.8	6.7±0.5	5.0±0.8	4.9±0.6

The predicted, central black hole masses shown here are in units of solar mass. As explained in Section 3, the predictions were derived from the galaxy's total stellar mass, spiral arm pitch angle and stellar velocity dispersion. The pitch angle, however, is considered unreliable for predicting the black hole mass in this post-merger galaxy, and it is therefore not used to calculate the error-weighted mean black hole mass, $\overline{\log M_{\mathrm{bh}}}$, shown in the final column. These masses should not be confused with the predicted, logarithmic black hole mass in Nikhuli, equal to 4.8 ± 1.6 (see the end of section 3).

3.2. *Nikhuli*

The central black hole of the infalling, nearly-consumed, galaxy might be smaller than the above prediction, and thus also of intermediate-mass (10^2–10^5 M$_\odot$). Graham (2020) has found that the M_{bh}–M_{nsc} relation for nuclear star clusters (NSCs) also applies to ultracompact dwarf (UCD) galaxies, thought to be the tidally stripped nuclei of galaxies (Zinnecker et al. 1988; Freeman 1993; Ferrarese et al. 2016). Nikhuli likely represents such a system. From a visual inspection of the 'residual image' of NGC 4424, it is hard to establish how much of the infalling nuclear star cluster may have been stripped away, or instead, how much of the infalling galaxy still remains around the infalling nuclear star cluster. Nonetheless, we can employ equation 7 from Graham (2020), which relates the stellar mass of nuclear star clusters to their central black hole mass. The relation can be written as

$$\log(M_{\mathrm{bh}}/\mathrm{M}_\odot) = (2.62 \pm 0.42)\log(M_{\mathrm{nc}}/[10^{7.83}\,\mathrm{M}_\odot]) + (8.22 \pm 0.20), \tag{3}$$

with the uncertainty on the black hole mass given by the expression

$$(\delta \log M_{\mathrm{bh}}/\mathrm{M}_\odot)^2 = [\log(M_{\mathrm{nc}}/[10^{7.83}\,\mathrm{M}_\odot])]^2(0.42)^2 + (0.20)^2 + (2.62)^2[\delta \log M_{\mathrm{nc}}]^2 + \epsilon^2.$$

[10] http://goldmine.mib.infn.it/

[11] This gives the 'maximum likelihood estimate' for the mean of the probability distributions under the assumption that they are independent and normally distributed with the same mean.

The term $\delta \log M_{\rm nc}$ is the uncertainty associated with the nuclear star cluster's stellar mass, and ϵ is the intrinsic scatter in the ($\log M_{\rm bh}$)-direction, and taken to be 1.31 dex (Graham 2020).

Assigning a factor of 2 uncertainty to the total stellar-mass of the stretched star cluster, one obtains a (poorly constrained)[12] value of $\log(M_{\rm bh}/{\rm M}_\odot) = 4.8 \pm 1.6$, corresponding to $M_{\rm bh} = 6 \times 10^4$ M$_\odot$ when expressed in linear units. For reference, due to the steep (2.62) slope of the $M_{\rm bh}$–$M_{\rm nsc}$ relation (Equation 3), a factor of 2 reduction to the star cluster mass results in a large 0.8 dex decrease to the predicted black hole mass in the star cluster. This black hole mass should not be confused with the above predictions for a *central* massive black hole in NGC 4424.

From $L_X \equiv L_{0.5-10\,{\rm keV}} = 7.09 \times 10^{38}$ erg s^{-1} (Table 1), the Eddington ratio ($L_X/L_{\rm Edd}$) of NGC 4424 X-3 can be calculated using $L_{\rm Edd} = 1.3 \times 10^{38} M_{\rm bh}/{\rm M}_\odot$ erg s^{-1}. Given the above black hole mass of 6×10^4 M$_\odot$, the Eddington ratio is 9×10^{-5}, or roughly 10^{-4}. Such a value is typical of those seen in the few active IMBH candidates at the centers of early-type galaxies in the Virgo cluster (Graham & Soria 2019, their Figure 8). Alternatively, NGC 4424 X-3 could be a 5.5 solar mass black hole radiating at the Eddington limit.

The likelihood of NGC 4424 X-3 being a compact stellar mass object, such as a neutron star or a stellar mass black hole, is low. High-mass X-ray binaries (HMXBs), in which the donor star has a high mass, obviously requires a young donor star. However, the morphology and red color of Nikhuli suggests that it consists of an older, evolved, stellar population, thereby disfavouring a HMXB. The X-ray luminosity function of low-mass X-ray binaries (LMXBs) scales with the stellar mass. From Lehmer et al. (2019), one expects around four LMXBs with $L_{0.5-8\,{\rm keV}} \geq 6 \times 10^{38}$ erg s^{-1} per 10^{11} solar masses, or 1.3×10^{-4} per 3.3×10^6 solar masses. That is, the expected number of LMXBs with $L_{0.5-8\,{\rm keV}} \geq 6 \times 10^{38}$ erg s^{-1} in Nikhuli is roughly 10^{-4}. In comparison, supermassive black holes and nuclear star clusters regularly coexist in galaxies, and the infall origin for Nikhuli does therefore favor the presence of a massive black hole. Confirmation of this logic and the mass of the black hole might be possible if broad optical emission lines are observed from NGC 4424 X-3. However, without the high spatial resolution offered by adaptive optics, such a signal from a somewhat weak IMBH may be lost in the noise of the galaxy light.

4. DISCUSSION

Although probably evident to many readers, for completeness, we note that Nikhuli has a lopsided shape, unlike a (background) galaxy The significant elongation of Nikhuli in the direction of the NGC 4424 galaxy center favours an infall scenario. Tidal (Roche lobe) stretching of an accreted nuclear star cluster, and possibly the paltry remains of the accreted galaxy still surrounding the nuclear star cluster, are evident. Lacking an outward-pointing, comet-like appearance, we suspect that this active[13] star cluster is probably not an ejected nucleus from a gravitational wave recoil event (Blecha & Loeb 2008; Holley-Bockelmann et al. 2008; Komossa & Merritt 2008; Askar et al. 2020; Hogg et al. 2021; Ward et al. 2021).

There is an abundance of nucleated dwarf and spiral galaxies (Böker et al. 2002; Graham & Guzmán 2003), and their nuclear star cluster can contain a massive black hole (e.g. Graham & Spitler 2009). The accretion of these galaxies, coupled with dynamical friction, can effectively drive their nuclear star clusters towards the centre of the accreting galaxy. This may result in massive black hole mergers and gravitational waves. Massive structures such as dense star clusters sink due to the braking force and thus orbital decay, arising from the dynamical friction with the surrounding galaxy (Chandrasekhar & von Neumann 1943; Baranov & Batrakov 1974; Tremaine et al. 1975; Inoue 2011; Arca-Sedda & Capuzzo-Dolcetta 2014; Chen et al. 2021; Morton et al. 2021). Differing from the *naked*[14] black holes in the merger simulations of Bellovary et al. (2021) and Ma et al. (2021), which tend not to sink to the center of the final galaxy, the *shrouded* black hole in Nikhuli should have more success (Pfister et al. 2019; Ma et al. 2021). Furthermore, the bar-like feature in NGC 4424 may further facilitate the inspiral of the Nikhuli/black hole system (Bortolas et al. 2021).

In low-mass spheroids with low Sérsic indices, and thus a shallow gradient to their central gravitational potential (e.g., Terzić & Graham 2005; Terzić & Sprague 2007), star clusters can wander about, typically displaced by 100 parsecs (Binggeli et al. 2000). Massive black holes can be similarly offset (Boldrini et al. 2020; Ricarte et al. 2021). A similar situation can occur in the partially depleted cores of massive spheroids, where infalling perturbers can stall outside of the centre (Read et al. 2006; Petts et al. 2015; Bonfini & Graham 2016; Banik & van den Bosch 2021). However, Figure 2 does not suggest that Nikhuli has stalled, and as such we can not conclude that Nikhuli is destined to become an eternally wandering black hole.

The phenomenon of an infalling galaxy undergoing assimilation has been offered to explain the offset IMBH in ESO 243-49 (Farrell et al. 2009; Bellovary et al. 2010; Mapelli et al. 2012; Webb et al. 2017; Bellovary et al.

[12] The heightened uncertainty is due to the steep slope of, and scatter about, the $M_{\rm bh}$–$M_{\rm nsc}$ relation.

[13] We use the term 'active' to denote a star cluster with an active black hole; in this case, X-ray active.

[14] The term 'naked' is used to denote the absence of a surrounding star cluster.

2021; Ricarte et al. 2021) and the ULX sources likely associated with a massive black hole in galaxies such as IC 4320 (Sutton et al. 2012), NGC 5252 (Kim et al. 2015, 2020) and others. One such interesting example can be seen in the Virgo cluster galaxy NGC 4651 (Martínez-Delgado et al. 2010; Foster et al. 2014; Morales et al. 2018), where the elongation process associated with infall has produced a tidal stream (Newberg 2016) which is immediately evident in optical images. An earlier stage of such shredding, of an infalling dwarf galaxy, has been reported in a galaxy at a redshift $z = 0.145$ by Forbes et al. (2003). See also UGC 10214 (Vorontsov-Velyaminov 1959; Miskolczi et al. 2011), NGC 7714 (Soria & Motch 2004), and M32 giving itself to M31 (Arp 1964; Graham 2002; Gordon et al. 2006). Various stages of the interaction process are evident in the catalog of Vorontsov-Velyaminov (1977), and the scenario in which a galaxy is threshed and reduced to its dense core has been developed by (Bekki & Freeman 2003).

In and around our Galaxy, we know that the Sagittarius dwarf galaxy is undergoing disruption (Ibata et al. 1994; Koposov et al. 2015), the Cetus and Indus stellar streams are the remnants of accreted dwarf galaxies Chang et al. (2020); Hansen et al. (2021), and the *Gaia-Enceladus-Sausage* was acquired long ago (Belokurov et al. 2018; Helmi et al. 2018). Furthermore, Ibata et al. (2019a) revealed the impressive tidal stream associated with ω Centauri, possibly the pared-core of an accreted dwarf galaxy but currently lacking confirmation of a central black hole (Zocchi et al. 2019; Baumgardt et al. 2019). Additional globular cluster streams, evidence of accretion and possibly remnant galaxy nuclei in the Milky Way are known (e.g. Bonaca et al. 2020; Woody & Schlaufman 2021; Pfeffer et al. 2020; Thomas et al. 2020; Ferguson et al. 2021; Piatti et al. 2021). Yuan et al. (2020) and Malhan et al. (2021) discovered and investigated the low-mass stream LMS-1, revealing the remnants of a low-mass galaxy now just 10 to 20 kpc from our Galaxy's center, making it the closest known stream to the Galaxy's center thus far. They conclude that the "globular cluster" NGC 5024 or NGC 5053 is likely the former nuclear star cluster of this low-mass galaxy, akin to NGC 5824 possibly being the remnant nuclear star cluster of the dwarf progenitor to the Cetus stream (Chang et al. 2020). The slightly more massive (than globular cluster) UCD galaxies (e.g. Ahn et al. 2017) have also been caught in the act of stripping. They are considered to be the former nuclei of galaxies (e.g., VUCD3: Liu et al. 2015) and are known to contain massive black holes (Graham 2020, and references therein).

The width of the Nikhuli stream is tens of parsecs to ∼100 pc across. As such, we are not seeing a captured globular cluster. Conceivably, the color or metallicity of Nikhuli might shed light on the full stellar mass of the progenitor via the color-magnitude or mass-metallicity relation for dwarf galaxies. However, given that the nuclear star cluster light will undoubtedly bias the flux, this would yield questionable results. With a stellar mass of $\sim 3\times10^6$ $M_\odot$, Nikhuli is more massive than globular clusters and thus more likely the nucleus of a stripped galaxy. Nonetheless, we note that globular clusters are known to contain XRBs (e.g. Hut et al. 1992; Becker et al. 2003; Maccarone et al. 2003), and it would therefore be desirable to observe Doppler-broadened, optical emission lines coming from NGC 4424 X-3 and thus verify the existence of the suspected massive black hole.

An incoming black hole of equal mass to that already (presumed to be) at the center of NGC 4424 might imply a major merger between two spiral galaxies. However, such a scenario is unlikely to result in a post-merger galaxy looking like a spiral galaxy. On the other hand, early-type galaxies have smaller stellar masses for a given black hole mass than late-type galaxies (Sahu et al. 2019). As such, a minor merger *can* bring in an equal mass black hole. This suggestion, which we think is presented here for the first time, may also explain why the extreme ULX and HLX off-center sources found by Sutton et al. (2012) in eight nearby (<100 Mpc) galaxies are not in galaxies displaying signs of a recent major merger.

A minor merger in NGC 4424 also meshes with the notion of an unequal (stellar mass) merger reported by Boselli et al. (2018). From Figure 11 in Sahu et al. (2019), one can see that an early-type galaxy with a black mass of $10^{4.8}$ $M_\odot$ (see section 3.2) will have a stellar mass of $\sim 0.6 \times 10^9$ $M_\odot$, which is just 6% of the stellar mass of NGC 4424 (see section 2.2). Suppose such an early-type galaxy is what fell into NGC 4424. In that case, (ignoring gaseous processes), the coalescence of NGC 4424's initial and incoming black holes will double the central black hole mass, with the stellar mass increasing by only ∼6%. We speculate that such minor mergers may help maintain the steepness of the non-linear, near-cubic $M_{\rm bh}$–$M_{*,\rm gal}$ relation for spiral galaxies (Savorgnan et al. 2016; Davis et al. 2018). Furthermore, compact spheroids and the bulges of early-type galaxies formed early in the Universe (Graham et al. 2015, and references therein) and were likely the formation sites of the first massive black holes. Some of these galaxies (and their black holes) may have subsequently been sewn (and sown) into late-type spiral galaxies.

Given the similarity of the predicted (central and infalling) black hole masses in NGC 4424 and the absence of an X-ray point source at the (very) center of NGC 4424, it might, but need not, be the case that NGC 4424 currently has no central black hole. The delivery of NGC 4424 X-3 via a low-mass early-type galaxy could be how bulgeless late-type galaxies are both seeded with black holes and commence building their bulge. In addition to a 'classical' (merger built) bulge, the incoming perturber may spark a bar instability leading to the creation of a larger scale (peanut shell)-shaped 'pseudobulge' structure (Combes et al. 1990; Athanassoula 2005;

Ciambur & Graham 2016). The median bulge-to-total flux ratios in Sc and Sd galaxies are only 8% (Graham & Worley 2008). At 6×10^{10} $M_\odot$ (Licquia & Newman 2015), the Milky Way has a stellar mass some six times bigger than NGC 4424. If six minor merger events have brought in six 10^5 $M_\odot$ black holes, one might (naively) expect a central black hole with a minimum mass of 0.6×10^6 $M_\odot$, with gas fuelling and stellar capture increasing this to the observed value of 4×10^6 $M_\odot$ (Boehle et al. 2016). It could also explain the tentative suggestions of IMBHs infalling/wandering in the Milky Way (Takekawa et al. 2020, and references therein).

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